Journal of Business Research 160 (2023) 113801



Contents lists available at ScienceDirect

Journal of Business Research

journal homepage: www.elsevier.com/locate/jbusres





The temperature of newness: How vision–temperature correspondence in advertising influences newness perception and product evaluation

Taku Togawa [a,*], Hiroaki Ishii [b], Jaewoo Park [c], Rajat Roy [d]

[a] Sophia University, Faculty of Economics, Department of Management, 7-1 Kioi-cho, Chiyoda-ku, Tokyo 102-8554, Japan
[b] Aoyama Gakuin University, School of Business, Department of Marketing, 4-4-25 Shibuya, Shibuya-ku, Tokyo 150-8366, Japan
[c] Chuo University, Faculty of Commerce, Department of Marketing and Trade, 742-1 Higashinakano Hachioji-shi, Tokyo 192-0393, Japan
[d] Associate Professor, Bond University, Bond Business School, Gold Coast Queensland 4226, Australia

ARTICLE INFO

Keywords:
Background ad imagery
Perceived temperature
Psychological distance
Ambiguity
Newness perception

ABSTRACT

Although studies have shown that background ad imagery can communicate various concepts to consumers, no research has investigated its effects on perceived ambiguity and newness of the advertised product. Drawing on construal level theory, we address this gap by exploring the psychological mechanism of newness perception. Through four studies, it is demonstrated that background ad imagery associated with coldness (vs. warmth) engenders perceptions of the newness of the advertised product (Study 1) as psychological distance enlarged by coldness enhances perceived ambiguity, a key driver of newness perception (Study 2). Moreover, the effect of background ad imagery on perceived newness manifests when the product style is modern (vs. antique) (Study 3) and subsequently improves product evaluation when consumers plan the purchase in the distant (vs. near) future (Study 4). These findings contribute to building a theoretical framework explaining how newness perception is communicated through advertising.

1. Introduction

Images placed on advertisements, namely advertising imagery, are a powerful tool to communicate desirable product concepts to consumers and create a strong position in a competitive market (Peracchio & Meyers-Levy, 2005). Previous studies have investigated the role of background ad imagery, which refers to advertising imagery depicting a non-product object placed behind or beside a product picture. For instance, luxury perception of an advertised product is enhanced by background ad imagery such as a photo of winter (vs. spring) scenery (Park & Hadi, 2020), vertical- (vs. horizontal-) oriented images (e.g., a picture of a skyscraper) (Van Rompay, De Vries, Bontekoe, & Tanja-Dijkstra, 2012), and a famous painting printed on the advertisement (Hagtvedt & Patrick, 2008). Furthermore, background ad imagery depicting nature is associated with the "greenness" of the advertised brand (Hartmann & Apaolaza-Ibáñez, 2009). While various perceptions have been comprehensively examined, less attention has been paid to how the perceived newness of the advertised product can be communicated to consumers (Goode, Dahl, & Moreau, 2013). We aim to address this gap by answering the following research question: How and when does background ad imagery convey newness of the product to consumers?

Perceived newness is defined as the extent to which consumers subjectively perceive a product as new, innovative, or creative (Goode et al., 2013). Previous research has highlighted the importance of newness perception by showing that it positively influences new product success (Blythe, 1999). The perception of product newness promotes consumers' adoption of the new product (Hetet, Ackermann, & Mathieu, 2020) and enhances its profitability (Calantone, Chan, & Cui, 2006). Furthermore, consumers indicate a higher willingness to pay for a product that they perceive as newer (Dahl & Moreau, 2002). In addition to these managerial implications, the literature has explored the psychological mechanisms by which consumers perceive a product as new (Goode et al., 2013). According to these findings, perceived newness tends to be engendered by consumers' perception of ambiguity, which refers to the difficulty in clearly assessing product features (Elsbach & Stigliani, 2019). To date, however, the factors that increase or decrease perceived ambiguity of an advertised product remain unclear (Cho & Taylor, 2020). It is theoretically important to address this gap because a sense of ambiguity or uncertainty has been considered a key driver affecting newness perception (Blake, Perloff, Zenhausern, & Heslin, 1973; Coskuner-Balli & Sandikci, 2014; Elsbach & Stigliani, 2019;

* Corresponding author.
E-mail addresses: togawa@sophia.ac.jp (T. Togawa), ishii@busi.aoyama.ac.jp (H. Ishii), jpark001@g.chuo-u.ac.jp (J. Park), rroy@bond.edu.au (R. Roy).

https://doi.org/10.1016/j.jbusres.2023.113801
Received 8 January 2022; Received in revised form 21 February 2023; Accepted 22 February 2023
Available online 1 March 2023
0148-2963/© 2023 Elsevier Inc. All rights reserved.

Goode et al., 2013).

Moreover, with a few exceptions (e.g., Hetet et al., 2020), research has not considered moderators determining when the newness perception is engendered and when it subsequently affects product evaluation. Studies have documented a congruence effect between temporality-related concepts (Biliciler, Raghunathan, & Ward, 2021; Chae & Hoegg, 2013); that is, when several temporal concepts are cognitively activated by factors such as ad imagery, product styles, and goal priming, congruence between the concepts (e.g., future-oriented goals and modern-style products) amplifies consumer perception and evaluation. Thus, perceived newness and its effect on product preferences can be affected when the temporal concept associated with newness (e.g., futuristic, advanced) matches the temporal concepts of other factors (e.g., product styles, purchase goals). To examine this congruence effect, we focus on two moderators, namely the temporality of product style (e.g., antique/modern) and the temporal distance to purchase (e.g., product purchase in the near/distant future).

Drawing on construal level theory (CLT), we predict that background ad imagery associated with coldness (vs. warmth) enhances perceived newness of the advertised product (Study 1). CLT posits that more psychologically distant objects are represented abstractly in people's minds (Liberman & Trope, 2008; Trope & Liberman, 2010). By applying this premise to the existing finding that the perception of coldness increases psychological distance (Ijzerman & Semin, 2009; Park & Hadi, 2020), we aim to demonstrate that psychological distance to an advertised product, enlarged by perceived coldness of background ad imagery, increases perceived ambiguity owing to abstract representations of the product. The heightened perception of ambiguity, in turn, enhances perceived newness of the advertised product (Study 2). Additionally, we identify a moderator—temporality of product style—by showing that the effect of background ad imagery on perceived newness manifests when the product style is modern and not antique (Study 3). Furthermore, we explore a downstream effect on product evaluation and a moderating role of temporal distance on the effect (Study 4). Specifically, we aim to demonstrate that when consumers plan to purchase a product in the distant (vs. near) future, perceived newness enhanced by background ad imagery improves (vs. impairs) product evaluation.

Building on CLT, our study contributes to the literature on advertising. Although previous studies have demonstrated that ad imagery affects consumers' psychological distance and construal levels (e.g., Kim, Lee, & Choi, 2019), they did not consider the role of temperature in this effect. Therefore, we aim to offer new insights by exploring how temperature, conveyed via background ad imagery, can determine psychological distance and influence abstractness of mental representations.

2. Theoretical background

2.1. CLT and psychological distance

CLT systematically explains the link between the psychological distance from an object and the way it is construed mentally (Liberman & Trope, 2008). According to this theory, objects, events, and individuals can be perceived as either psychologically near or far, along different dimensions of distance-spatial, temporal, social, and hypothetical, for example (Liberman & Trope, 2008; Trope, Liberman, & Wakslak, 2007). CLT posits that people tend to use concrete and low-level construal to represent psychologically near objects, and abstract and high-level construal to represent psychologically distant objects (Fujita, Eyal, Chaiken, Trope, & Liberman, 2008; Trope et al., 2007; Trope & Liberman, 2003). For instance, when viewing actions (e.g., "reading a book") by socially distant others, people tend to describe them with abstract versus concrete representations (e.g., "gaining knowledge" rather than "following lines of print") (Liviatan, Trope, & Liberman, 2008). Similarly, when thinking about a temporally distant (vs. near) goal, people prefer abstract (vs. concrete) representations of the goal (Liberman & Trope, 1998; Sánchez, Coleman, & Ledgerwood, 2021). Applying the premise of the theory to the marketing context, previous research has also demonstrated that when consumers imagine a temporally distant (vs. near) purchase, they construe the purchase abstractly and make a judgment about products relying on abstract (vs. concrete) information (Roehm & Roehm, 2010; Yan & Sengupta, 2011).

In line with the notion of CLT, the following sections outline our key premise that background ad imagery, related to coldness, enhances perceived newness of the advertised product.

2.2. Vision–Temperature correspondence and newness perception

An object's temperature is often perceived through visual cues rather than physical touch (Ernst & Banks, 2002; Szocs & Biswas, 2013); for example, people feel cold on seeing (but not touching) a cold object (e.g., seeing a piece of ice versus a transparent plastic cube placed in someone's hand) (Kanaya, Matsushima, & Yokosawa, 2012). Empirical evidence for this "vision–temperature correspondence" has been reported in marketing literature as well. Park and Hadi (2020) examined the effects of perceived coldness (vs. warmth) on consumers' perception of a product's luxuriousness, which manipulates perceived temperature not only by altering physical temperature but also by presenting a picture depicting either spring or winter scenery. Similarly, in another study, images of cold objects (e.g., ice, frost, snow) led to feelings of physical coldness and loneliness (Choi, Rangan, & Singh, 2016).

We focus on the role of perceived temperature in psychological distance to an object. Previous studies relying on grounded cognition theory posit that the experience of physical warmth activates metaphorical concepts related to spatial proximity, as warmth is often associated with interpersonal proximity and intimacy (Barsalou, 2010; Landau, Meier, & Keefer, 2010). As spatial distance can determine other subdimensions of psychological distance (e.g., social, temporal) (Zhang & Wang, 2009), the activated concept of spatial proximity influences social distance from others. For example, the experience of physical warmth increases feelings of interpersonal warmth (Williams & Bargh, 2008) and decreases the social distance to others (Ijzerman & Semin, 2009). Conversely, cold temperatures increase social distance and exclusion perceptions (Ijzerman & Semin, 2009; Park & Hadi, 2020). Based on these findings, we assume that perceived coldness of background ad imagery increases the psychological distance to the advertised product.

As noted, psychological distance includes a subdimension of hypothetical distance, which refers to the extent to which the occurrence of an event, existence of an object, or degree of closeness to an individual is probable in real life (Trope et al., 2007; Trope & Liberman, 2010). Therefore, when psychological distance to the advertised product is enlarged by background ad imagery, it seems to consumers that the product does not exist in reality. In general, people have limited information about objects with which they cannot interact directly in the real world (Liberman & Trope, 2008). Previous studies suggest that a lack of concrete information increases perceived ambiguity, defined as the potential for multiple interpretations of product features (Hoch & Ha, 1986). Based on these arguments, we postulate that background ad imagery related to coldness increases psychological distance to the advertised product, which in turn increases the perception of ambiguity.

Several studies suggest that ambiguity is the key factor determining newness perception. For instance, a qualitative study by Elsbach and Stigliani (2019) shows that people generally believe that new information technology is "mysterious," "ambiguous," and "unfamiliar." A quantitative study demonstrated that consumers who tended to seek ambiguity preferred a product that was perceived as new, suggesting that ambiguity serves as a determinant of newness perception (Blake et al., 1973). Moreover, newness perception increases when the point of reference to assess the new product is ambiguous or difficult to determine (Goode et al., 2013). Therefore, we predict that feelings of ambiguity, driven by the background ad imagery via psychological distance,

will enhance the perceived newness of the product. More formally, we hypothesize the following:

H1a: Background ad imagery associated with coldness (vs. warmth) enhances the perceived newness of the advertised product.
H1b: The effect predicted in H1a is sequentially mediated by the psychological distance and ambiguity of the product.

2.3. Moderating role of temporality represented by product style

To explore a possible moderator of the effect predicted in H1a, we focus on the role of temporality represented by product style. Previous studies have shown that product style (e.g., shape, color) represents specific concepts related to temporality (Bloch, 1995; Creusen & Schoormans, 2005; Noble & Kumar, 2010). For instance, antique product styles represent the concept of "the past," whereas modern product styles communicate the concept of "the future" (Chae & Hoegg, 2013). We assume that antique and modern styles are associated with temporal proximity and remoteness, respectively, based on the subsequent rationale.

Previous studies have shown that past events tend to be described with more concrete words than future events (Johnson, Foley, Suengas, & Raye, 1988; Snefjella & Kuperman, 2015). As the relationship between construal level and psychological distance is bidirectional (i.e., concretely construed events are perceived as closer) (Van Boven, Kane, McGraw, & Dale, 2010), past- (vs. future-) related objects that are construed with more concrete (vs. abstract) representations will be associated with temporal closeness (vs. distance). Therefore, we predict that the perception of temporal distance can be manipulated by the temporality of product style; that is, people will associate an antique- (vs. modern-) style product with temporal proximity (vs. remoteness).

Considering the association between the past and temporal proximity, this link will moderate the effect of background ad imagery on perceived newness of the advertised product. As discussed, our basic prediction (H1a) is based on the notion that psychological distance enlarged by perceived coldness of background ad imagery increases the sense of ambiguity and perceived newness of the product. Therefore, this effect will be canceled out by the temporal proximity of the product with an antique style. In contrast, given that people associate a modern product style with temporal remoteness, this perception of distance fits with the psychological distance to the product, which is highlighted by the perceived coldness of background ad imagery. Moreover, as consumers' responses tend to be bolstered by a congruence between several dimensions of psychological distance (Kim, Zhang, & Li, 2008; Zhao & Xie, 2011), temporal remoteness of the modern-style product will promote the effect of perceived coldness of background ad imagery on newness perception. Thus, the following hypothesis is deduced:

H2: The effect of background ad imagery related to coldness on perceived newness (H1a) is moderated by temporality represented by product style. Specifically, the effect manifests when the product style is modern and not antique.

2.4. Downstream effect on product evaluation

How and when does newness perception driven by background ad imagery subsequently influence product evaluation? Previous studies suggest that consumers' preference for new, innovative products can vary depending on various moderators, such as consumers' psychological characteristics (Blythe, 1999; Ma, Yang, & Mourali, 2014) and product types (Schnurr, 2017). Temporal distance to purchase may also be one of the moderators, given that construal levels moderate the effect of specific perceptions (e.g., aesthetic perception) evoked by ad imagery on the evaluation of the advertised product (Castagna, Pinto, Mattila, & de Barcellos, 2021). Thus, relying on CLT, we further explore the moderating role of temporal distance in the effect of newness perception on product evaluation.

A body of CLT literature commonly suggests that when a temporally distant (vs. near) object is primed, more abstract (vs. concrete) information increases preference for the object (Fujita et al., 2008; Trope & Liberman, 2000, 2010). For example, monochrome (vs. color) ad imagery lacks concrete color information and induces abstract mindset, resulting in a higher preference for distant (vs. near) future events (Lee, Fujita, Deng, & Unnava, 2017). As outlined in the preceding section, perceived newness is magnified by the ambiguity of the product features owing to a lack of concrete information (Elsbach & Stigliani, 2019; Goode et al., 2013); this suggests that products perceived as new are more likely to be represented abstractly by consumers. Therefore, when consumers plan a product purchase in the distant future, they will prefer a product that is perceived as newer.

Conversely, given that a product perceived as new is represented more abstractly, the representations will not be compatible with temporal proximity to the product purchase. This incongruence between temporal distance and abstractness of the representations will negatively influence product evaluation (Fujita et al., 2008; Liberman, Sagristano, & Trope, 2002; Trope & Liberman, 2000). Therefore, we propose the following hypothesis:

H3: The effect of perceived newness engendered by background ad imagery on product evaluation is moderated by temporal distance to purchase. When planning a temporally distant purchase, newness perception improves evaluation of the advertised product. However, when consumers imagine a temporally near purchase, newness perception evoked by background ad imagery negatively impacts evaluation of the advertised product.

3. Overview of studies

We tested H1a, H1b, H2, and H3 through four experiments. The overall framework and the studies are presented in Fig. 1. After investigating the main effect of coldness-related background ad imagery on newness perception (H1a) in Study 1, we explored the underlying mechanism of this effect by running a serial mediation analysis with psychological distance and ambiguity as mediators (H1b) in Study 2. In Study 3, we investigated whether the basic effect predicted in H1a was moderated by temporality represented by product style (i.e., "antique" vs. "modern") (H2). Finally, in Study 4, we tested H3 by examining downstream effects on product evaluation considering consumers' temporal distance (near/distant) as a moderator.

All studies were conducted in Japan from spring to autumn using an online survey platform. Participants were recruited using CrowdWorks, one of the major crowdsourcing platforms in Japan (see Online Supplemental Material for more details). We manipulated visual perceptions of temperature using background ad imagery placed behind (Studies 1 and 2) or beside (Studies 3 and 4) a product picture. By using different types of stimuli, we attempted to ensure the generalizability of our findings. Across the experiments, all variables were measured using a seven-point scale (1 = strongly disagree to 7 = strongly agree), except as specified. The measurement items used in the experiments are summarized in Table 1.

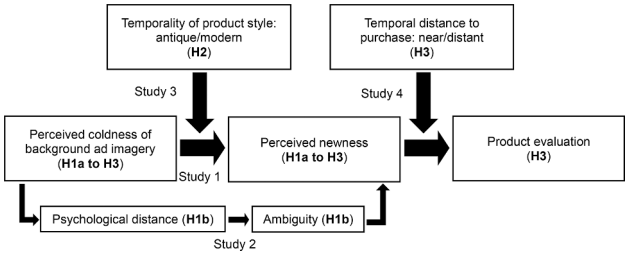


Fig. 1. Overall theoretical framework and studies.

Table 1
Measurement items used in the studies.

Study	Variable	Item	Source
Study 1 to 4	Perceived temperature of the background imagery	1 = "Cold" to 7 = "Hot" (bipolar scale)	Park and Hadi (2020)
Study 1 to 4	Liking of the scenery image	"Likable," "Good," "Favorable"	Aaker (2000)
Study 1 to 4	Perceived newness of the advertised product	"New," "Novel," "Innovative," "Creative"	Goode et al. (2013)
Study 1 to 4	Felt temperature of the participants' physical environment	1 = "Cold" to 7 = "Hot" (bipolar scale)	Park and Hadi (2020)
Study 1 to 4	Device type	0 = "Smart devices such as smartphones and tablets/1 = "Personal computers such as laptops and desktop computers"	Original
Study 1 to 4	Perceived typicality of the background imagery	"This background imagery is typical." "This background imagery is unrepresentative (reverse-coded)."	Loken and Ward (1990)
Study 2	Visual clarity	"Clear," "Sharp"	Hagtvedt (2011)
Studies 2 and 4	Freshness	"I felt the product was fresh."	Gvili et al. (2017)
Studies 2 and 4	Psychological distance	"I felt the product was far away."	Zhang and Wang (2009)
Studies 2 and 4	Ambiguity	"I was not sure about the product." "I struggled to evaluate the product."	Cho and Taylor (2020)
Studies 2 and 4	Positive mood	"Lively," "Glad," "Active," "Engrossed," "Friendliness," "Passionate," "Enthusiastic," "Fun," "Determined," "Pleasure," "Delighted"	Clark and Watson (1989)
Studies 2 and 4	Negative mood	"Embarrassed," "Suffering," "Discouraged," "Nervous," "Angry," "Feel inferior," "Disappointed," "Shaking," "Flustered," "Self-doubt"	Clark and Watson (1989)
Study 4	Car's trait (driving performance)	"This car would have a superior driving performance."	Original
Study 4	Car's trait (durability)	"This car would be durable."	Original
Study 4	Temporal distance	1 = "Very close in time" to 7 = "Very distant in time"	Wohl and McGrath (2007)
Study 4	Product evaluation	"Likable," "Good," "Favorable"	Aaker (2000)

Note. The items were measured on a seven-point scale (1 = "Strongly disagree" to 7 = "Strongly agree"), except when specified. Besides these items, participants completed the questions for attention check and demographic information in each study.

As it was conducted online, it was not possible to control the actual temperature of the place where participants completed the survey. Therefore, we measured the felt temperature of the participants' physical environment through a question before starting the main part of the experiments. This enabled us to control for this factor as a covariate in the analysis of covariance; the results of the analysis are reported in the Online Supplemental Material.

To address concerns about the data quality of online surveys, we used attention check questions throughout the studies and excluded from analysis participants who did not pay sufficient attention during the experiment. More detailed information about the attention check and exclusion criteria is disclosed in the Online Supplemental Material.

4. Study 1

In Study 1, we aimed to examine whether background ad imagery associated with coldness (vs. warmth) enhances the newness perception of the advertised product (H1a).

Prior to the main study, we created a fictitious advertisement for sneakers as experimental stimuli (see Appendix A). Previous studies have demonstrated that consumers associate specific types of color (e.g., blue) with innovativeness and creativity compared with other colors (e.g., yellow) (Kim, Hyun, & Park, 2020). To control for this effect, we employed black and white stimuli.

4.1. Methods

In total, 154 participants (M_{age} = 38.98, SD_{age} = 8.80; 95 women) were randomly assigned to either the warm or cold condition. First, participants were asked to indicate the temperature at the place from which they were completing the survey, to account for this covariate. Next, we presented an image of a pair of sneakers placed on a wooden floor (warm condition) or crushed ice (cold condition). Then, participants rated the perceived temperature of the background ad imagery (1 = Cold to 7 = Hot) (Park & Hadi, 2020) for the manipulation check. Furthermore, we measured liking of the background using three options (e.g., "good," "favorable") for the confound check, followed by perceived newness of the sneakers with four options (e.g., "new," "novel") (Goode et al., 2013) for a key dependent variable, averaged to form the indices of background liking ($\alpha = 0.93$) and perceived newness ($\alpha = 0.92$), respectively.

The perceived typicality of background ad imagery may influence the perceived newness of the advertised product, given that relevant concepts, such as the product's novelty and creativity, are components of consumers' perception of a product's newness and innovativeness (Goode et al., 2013; Holbrook, 1981). To rule out this explanation, we also measured the perceived typicality of the background using two items (e.g., "This background imagery is typical") (Loken & Ward, 1990); both items were averaged to form the typicality index ($r = 0.52$). Finally, participants completed two attention check questions; provided demographic information, such as gender and age; and specified the device type used to complete the study.

4.2. Results and discussion

Based on the responses to the attention check questions, three participants were excluded from the analysis (final N = 151).

Perceived temperature in the cold condition was significantly lower than that in the warm condition ($M_{warm} = 2.92$, $SD_{warm} = 0.85$ vs. $M_{cold} = 2.21$, $SD_{cold} = 0.84$; $F(1, 149) = 26.55$, $p < .01$, $\eta_p^2 = .15$), whereas there was no significant difference between warm and cold conditions in both background liking ($M_{warm} = 4.46$, $SD_{warm} = 1.14$ vs. $M_{cold} = 4.61$, $SD_{cold} = 1.10$; $F(1, 149) = 0.66$, $p = .42$, $\eta_p^2 < .01$) and background typicality ($M_{warm} = 4.39$, $SD_{warm} = 1.13$ vs. $M_{cold} = 4.27$, $SD_{cold} = 1.02$; $F(1, 149) = 0.48$, $p = .49$, $\eta_p^2 < .01$). Thus, the manipulation of visual coldness perception was successful, and the confounders remained constant. More importantly, participants in the cold condition perceived the pair of sneakers as newer than did those in the warm condition ($M_{warm} = 3.23$, $SD_{warm} = 1.24$ vs. $M_{cold} = 3.66$, $SD_{cold} = 1.145$; $F(1, 149) = 4.87$, $p = .03$, $\eta_p^2 = .03$). Thus, in Study 1, we found that background ad imagery associated with coldness (vs. warmth) enhanced the perceived newness of the product, providing initial evidence supporting H1a.

We also conducted a supplementary study that included a control condition as well as cold and warm conditions (see Online Supplemental Material for details). In the cold and warm conditions, an image of the advertised product (i.e., smartwatch) was placed against the background depicting winter and spring scenery, respectively. The stimulus for the control condition did not have any background imagery; that is, an image of the smartwatch was placed against a white background. Participants in the cold condition indicated a stronger perception of the product's newness than did those in the warm condition (adjusted p =

.01) and the control condition (adjusted p = .04). Meanwhile, perceived newness in the warm condition was not significantly different from that in the control condition (adjusted p = 1.00). Thus, for methodological simplicity, the following experiments only focused on comparisons between warm and cold conditions.

5. Study 2

Study 2 was designed to identify the underlying mechanism of the effect of background ad imagery associated with coldness on the perceived newness of the product (H1b), and to rule out alternative explanations for the underlying mechanism. First, previous studies suggest that positive or negative mood, which can be generated by perceived temperature (Howarth & Hoffman, 1984), may influence the perceived newness of the product (Bartels & Reinders, 2011). Specifically, there is the possibility that the negative mood generated by the perception of coldness enhances a participant's creative mindset (George & Zhou, 2002), in turn making them seek and expect a product's novelty (Hirschman, 1980).

Second, in everyday life, people experience and learn that cold food products are generally fresh (Roque, Auvray, & Lafraire, 2018). Given that consumers strongly associate coldness with freshness, the link activated by background ad imagery may enhance the newness perception of the advertised product. In this study, we attempted to rule out these explanations by measuring and analyzing participants' mood status and perceived freshness.

5.1. Method

5.1.1. Design and participants

A two-cell, between-participant design was employed in this study. As with the previous experiments, we used different types of background ad imagery to manipulate visual perception of temperature. In total, 172 participants (M_{age} = 41.27, SD_{age} = 10.35; 101 women) were randomly assigned to either the warm or cold condition.

5.1.2. Stimuli and procedure

After measuring the felt temperature of participants' present environment using the same scale as in the previous experiments, we presented the same stimulus as in the supplementary study-an advertisement for a smartwatch with a background image depicting either spring (warm condition) or winter scenery (cold condition) (see Appendix B).

5.1.3. Measures

Participants rated perceived temperature (single item) and liking (α = 0.95) of the background imagery, typicality of the background (r = 0.44), and perceived newness of the product (α = 0.89) on the same scales used in the previous experiments. To confirm that the product could be seen clearly regardless of the background imagery, participants also evaluated visual clarity on two items ("clear" and "sharp"; r = 0.85), averaged to form a visual clarity index. Additionally, we measured psychological distance to the product ("I felt the product was far away") (Zhang & Wang, 2009) and ambiguity of the product (e.g., "I was not sure about the product"; r = 0.45) (Cho & Taylor, 2020) as mediators. Participants' responses to the two items for ambiguity were averaged to form the ambiguity index. To examine alternative explanations, we also measured perceived freshness using a single item ("I felt the product was fresh") (Gvili, Tal, Amar, & Wansink, 2017) and their current mood status on the translated Japanese version of the Positive and Negative Affect Schedule (Clark & Watson, 1989). In the analysis, the items for positive and negative mood were averaged separately to form the positive (α = 0.94) and negative (α = 0.95) mood indices.

Finally, participants completed attention check questions, provided demographic information, and clarified the device type used to complete the study.

5.2. Results and discussion

Based on the results of the attention check questions, two participants were excluded from the following analysis (final N = 170).

5.2.1. Manipulation check

Perceived temperature in the cold condition was significantly lower than that in the warm condition (M_{warm} = 4.35, SD_{warm} = 1.00 vs. M_{cold} = 1.95, SD_{cold} = 0.69; F (1, 168) = 244.04, p < .01, η_p^2 = .66), whereas there were non-significant differences between the warm and cold conditions in liking of the background imagery (M_{warm} = 4.27, SD_{warm} = 1.36 vs. M_{cold} = 4.09, SD_{cold} = 1.38; F (1, 168) = 0.73, p = .39, η_p^2 < .01), perceived typicality of the background imagery (M_{warm} = 4.01, SD_{warm} = 1.01 vs. M_{cold} = 3.77, SD_{cold} = 1.08; F (1, 168) = 2.20, p = .14, η_p^2 = .01), and visual clarity of the product (M_{warm} = 4.54, SD_{warm} = 1.19 vs. M_{cold} = 4.52, SD_{cold} = 1.13; F (1, 168) = 0.01, p = .93, η_p^2 < .01), suggesting that the manipulation of perceived temperature was successful.

5.2.2. Psychological distance, ambiguity, and perceived newness

In the cold condition, the product (i.e., smartwatch) was perceived as more psychologically distant and ambiguous compared to the warm condition (psychological distance: M_{warm} = 3.77, SD_{warm} = 1.39 vs. M_{cold} = 4.48, SD_{cold} = 1.37; F (1, 168) = 11.27, p < .01, η_p^2 = .06; ambiguity: M_{warm} = 4.59, SD_{warm} = 1.09 vs. M_{cold} = 3.92, SD_{cold} = 1.21; F (1, 168) = 14.13, p < .01, η_p^2 = .08). Moreover, participants in the cold condition perceived the product as newer than did those in the warm condition (M_{warm} = 4.09, SD_{warm} = 1.25 vs. M_{cold} = 4.60, SD_{cold} = 1.14; F (1, 168) = 7.72, p = .01, η_p^2 = .04), replicating the results for H1a.

5.2.3. Underlying mechanism

We performed a serial mediation analysis with background ad imagery (0: warm, 1: cold; dummy coded) as the independent variable, perceived newness as the dependent variable, and psychological distance and ambiguity as the mediators, using PROCESS Model 6 (Hayes, 2021) (Fig. 2). The analysis revealed significant relationships between background ad imagery and psychological distance (B = 0.71, SE = 0.21, t = 3.36, p < .01), psychological distance and ambiguity (B = 0.27, SE = 0.06, t = 4.34, p < .01), and ambiguity and perceived newness (B = 0.22, SE = 0.08, t = 2.61, p = .01). The effect of background ad imagery on ambiguity was also significant (B = 0.46, SE = 0.17, t = 2.74, p = .01), whereas the relationship between psychological distance and perceived newness was non-significant (B = 0.01, SE = 0.07, t = 0.17, p = .87).

Moreover, bootstrapping with 10,000 resamples revealed a significant indirect effect of psychological distance and ambiguity on perceived newness (index = 0.04, SE = 0.02, 95% CI [0.01, 0.10]), supporting H1b. Additionally, a serial mediation path with the reversed order of the two mediators was non-significant (index < 0.01, SE = 0.02, 95% CI [-0.04, 0.04]), suggesting that these mediators were not interchangeable.

5.2.4. Alternative explanations

Furthermore, there were non-significant differences, both in the positive mood index (M_{warm} = 3.42, SD_{warm} = 1.33 vs. M_{cold} = 3.24, SD_{cold} = 0.97; F (1, 168) = 1.06, p = .30, η_p^2 = .01) and the negative mood index (M_{warm} = 2.63, SD_{warm} = 1.27 vs. M_{cold} = 2.81, SD_{cold} = 1.33; F (1, 168) = 0.79, p = .38, η_p^2 = .01), ruling out the possibility that specific types of mood generated by perceived temperature influenced participants' evaluation of the product's newness. Additionally, there was a non-significant difference in perceived freshness between the warm and cold conditions (M_{warm} = 4.02, SD_{warm} = 1.38 vs. M_{cold} = 3.81, SD_{cold} = 1.33; F (1, 168) = 1.06, p = .31, η_p^2 = .01). Thus, the alternative account based on the perception of freshness was also ruled out.

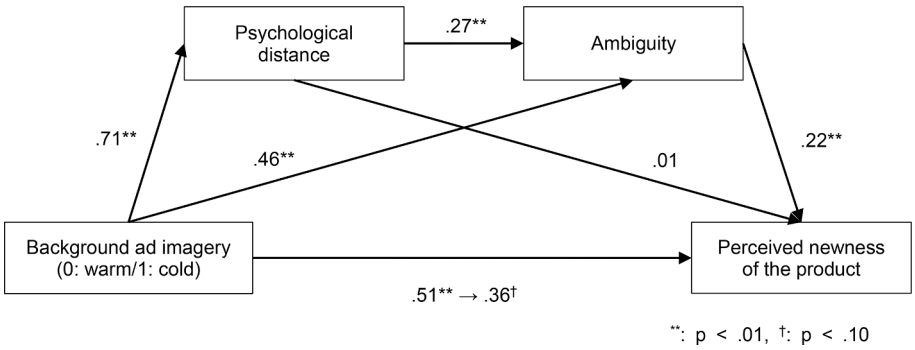


Fig. 2. Psychological distance and product ambiguity mediated the relationship between background ad imagery and perceived newness of the advertised product. Numbers in the figure represent unstandardized coefficients.

6. Study 3

Study 3 was designed to test whether temporality (i.e., antique vs. modern), represented by product style, moderates the effect of background ad imagery on newness perceptions (H2).

6.1. Methods

6.1.1. Design and participants

A 2 (background ad imagery: warm/cold) × 2 (temporality of product style: antique/modern) between-participants design was employed. In total, 218 participants (M_{age} = 38.96, SD_{age} = 10.46, 117 women, and two non-responses to the question on gender) were randomly assigned to one of the four conditions.

6.1.2. Stimuli

To manipulate the temporality represented by product style, we created fictitious advertisements for antique-style and modern-style table lamps. Additionally, the advertisements included a photo depicting either spring (warm condition) or winter (cold condition) scenery to manipulate participants' perception of temperature (see Appendix C). A pretest (N = 102) confirmed that the modern- (vs. antique-) style lamp was perceived as modern (vs. antique). The two lamps did not differ according to participants' evaluation of luxuriousness and preference (see the Online Supplemental Material for the pretests).

6.1.3. Procedure

We presented participants with a short description: "This is an advertisement for a table lamp produced by a popular furniture maker," followed by one of the stimuli. Participants rated perceived temperature, liking for the scenery photo (α = 0.95), typicality of the background ad imagery (r = 0.50), and perceived newness of the product (α = 0.87) using the same items as in Study 1. Finally, they responded to the attention check questions, mentioned the device type they used to complete the study, and provided demographic information.

6.2. Results and discussion

Based on the results of the attention check questions, seven participants were excluded from the analysis (final N = 211).

6.2.1. Manipulation check

Consistent with the pretest, a one-way ANOVA showed that the perceived temperature index in the cold condition was significantly lower than that in the warm condition (M_{warm} = 4.41, SD_{warm} = 1.09 vs. M_{cold} = 2.91, SD_{cold} = 1.44; F (1, 209) = 73.13, p < .01, η_p^2 = .26), whereas there were non-significant differences between the two temperature conditions in both liking for the scenery photo (M_{warm} = 4.35, SD_{warm} = 1.43 vs. M_{cold} = 4.37, SD_{cold} = 1.43; F (1, 209) < 0.01, p = .94, η_p^2 < .01) and typicality of the background imagery (M_{warm} = 4.12, SD_{warm} = 1.27 vs. M_{cold} = 4.24, SD_{cold} = 1.00; F (1, 209) = 0.61, p = .44, η_p^2 < .01). Therefore, we confirmed that the manipulation of perceived temperature was successful.

6.2.2. Perceived newness

A 2 (background ad imagery: warm/cold) × 2 (product style: antique/modern) ANOVA revealed a significant main effect of background ad imagery on perceived newness (M_{warm} = 3.35, SD_{warm} = 1.09 vs. M_{cold} = 3.51, SD_{cold} = 1.15; F (1, 207) = 5.04, p = .03, η_p^2 = .02), replicating the results for H1a. The main effect of product style on perceived newness was also significant ($M_{antique}$ = 3.11, $SD_{antique}$ = 1.03 vs. M_{modern} = 3.88, SD_{modern} = 1.10; F (1, 207) = 27.09, p < .01, η_p^2 = .12). Furthermore, the analysis showed a significant interaction between background ad imagery and product style (F (1, 207) = 7.26, p = .01, η_p^2 = .03) (Fig. 3). For the modern condition, participants in the cold condition perceived the table lamp as newer than did those in the warm condition (M_{warm} = 3.51, SD_{warm} = 1.13 vs. M_{cold} = 4.24, SD_{cold} = 0.94; F (1, 207) = 11.50, p < .01, η_p^2 = .05). In the antique condition, however, background ad imagery had no significant impact on perceived newness (M_{warm} = 3.14, SD_{warm} = 1.01 vs. M_{cold} = 3.08, SD_{cold} = 1.05; F (1, 207) = 0.11, p = .74, η_p^2 < .01). These results support H2.

7. Study 4

In Study 4, we examined the downstream effect of newness perception, engendered by background ad imagery, on product evaluation. As predicted in H3, we expected temporal distance to moderate the effect of newness perception on product evaluation; that is, when planning a product purchase in the distant (vs. near) future, perceived newness enhanced by background ad imagery would improve (vs. impair) the evaluation of the advertised product. We tested this prediction using a moderated mediation analysis.

7.1. Method

7.1.1. Design and participants

A 2 (background ad imagery: warm/cold) × 2 (temporal distance: near/distant) between-participants design was employed. As with the previous experiments, two types of background imagery (warmth- and coldness-related images) were used to manipulate perceived temperature. Moreover, we manipulated temporal distance to purchase by asking participants to imagine a situation where they purchase a product in the distant or near future. In the study, 221 participants (M_{age} = 40.66, SD_{age} = 9.14; 111 women, and one non-response to the gender question) were randomly assigned to one of the four conditions.

7.1.2. Stimuli

We created fictitious advertisements for a sedan car as the

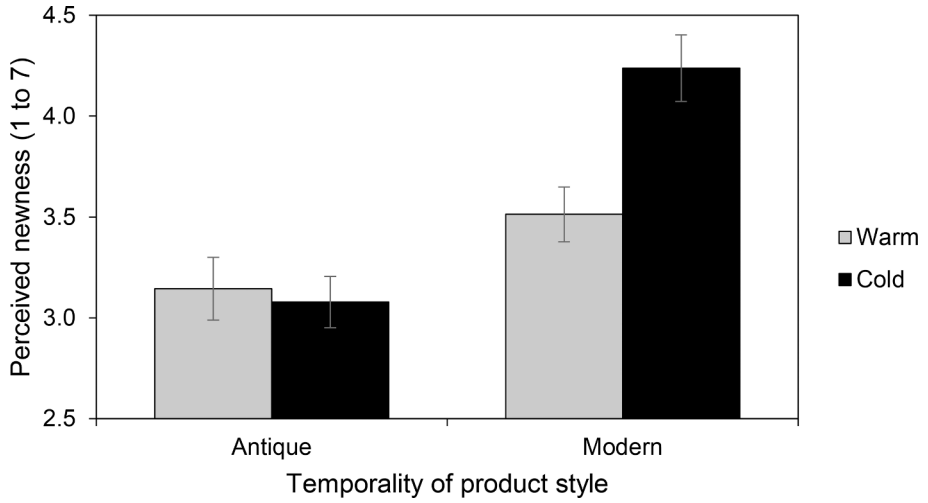


Fig. 3. Effects of background ad imagery on perceived newness as a function of temporality represented by product style. Error bars represent standard error (±1 SE).

experimental stimuli. Similar to Studies 2 and 3 and the supplementary study, we manipulated visual temperature perception by using images of spring or winter scenery in the advertisements (see Appendix D). The pretest for the supplementary study had confirmed that people perceived the image with winter scenery as colder than the spring scenery, whereas there was no significant difference in participants' liking for the scenery photos (see the Online Supplemental Material for details).

7.1.3. Procedure

Prior to the main part of the study, participants indicated the temperature of their environment, using the same scale as in the previous experiments.

In the main part of the study, to manipulate temporal distance to purchase, participants were instructed to read one of the following scenarios (italics added here for emphasis): "Please imagine that you plan to purchase a new car *one month later* [near-future condition]*/one year later* [distant-future condition]. One day, you happen to see the following advertisement for a car." For the manipulation check of temporal distance, we asked participants to indicate how distant the purchase felt to them (1 = "Very close in time" to 7 = "Very distant in time") (Wohl & McGrath, 2007).

Next, the participants were shown an advertisement with a picture of natural scenery in either spring (warm condition) or winter (cold condition), and asked to rate the perceived temperature of the scenery photo, perceived freshness, liking for the scenery photo ($\alpha = 0.88$), background typicality ($r = 0.49$), and positive ($\alpha = 0.93$) and negative ($\alpha = 0.95$) mood, using the same scales as in the previous experiments. A possible confound was that the spring or winter scenery photos in the stimuli could produce different interpretations of the product's traits; for example, participants could associate a snow scene with the car's driving performance. To check this confounding factor, we asked participants to rate the car's traits associated with the scenery photo using two scales: driving performance and durability (Table 1).

Following the ad evaluation task, we measured product evaluation using three items (e.g., "favorable") (Aaker, 2000), averaged to form the product evaluation index ($\alpha = 0.93$). For replication, we also measured psychological distance to the product, product ambiguity ($r = 0.47$), and perceived newness of the product ($\alpha = 0.87$) on the same scales as in the previous experiments. At the end of the study, participants completed two attention check questions, mentioned the device type they used to complete the study, and provided demographic information.

7.2. Results and discussion

None of the participants met our pre-defined exclusion criteria; thus, all participants' responses were included in the following analyses.

7.2.1. Manipulation check

Participants in the distant-future condition felt that the purchase described in the scenario was more temporally distant than did those in the near-future condition ($M_{near} = 4.10$, $SD_{near} = 1.51$ vs. $M_{distant} = 5.32$, $SD_{distant} = 1.62$; $F(1, 219) = 33.23$, $p < .01$, $\eta_p^2 = .13$). Thus, the manipulation of temporal distance was successful.

Additionally, the perceived temperature of the picture depicting winter scenery was significantly lower than that of the picture depicting spring scenery ($M_{warm} = 4.80$, $SD_{warm} = 1.16$ vs. $M_{cold} = 1.87$, $SD_{cold} = 0.84$; $F(1, 219) = 468.12$, $p < .01$, $\eta_p^2 = .68$). There was no significant difference between the warm and cold conditions in liking for the scenery photos ($M_{warm} = 4.99$, $SD_{warm} = 1.20$ vs. $M_{cold} = 4.83$, $SD_{cold} = 0.89$; $F(1, 219) = 1.17$, $p = .28$, $\eta_p^2 = .01$) and typicality of the background imagery ($M_{warm} = 4.83$, $SD_{warm} = 1.17$ vs. $M_{cold} = 4.66$, $SD_{cold} = 1.03$; $F(1, 219) = 1.24$, $p = .27$, $\eta_p^2 = .01$). Moreover, the scenery photos were not significantly associated with certain product traits, namely driving performance ($M_{warm} = 3.72$, $SD_{warm} = 1.26$ vs. $M_{cold} = 3.81$, $SD_{cold} = 1.36$; $F(1, 219) = 0.27$, $p = .60$, $\eta_p^2 < .01$) and durability ($M_{warm} = 3.80$, $SD_{warm} = 1.17$ vs. $M_{cold} = 4.03$, $SD_{cold} = 1.31$; $F(1, 219) = 1.78$, $p = .18$, $\eta_p^2 = .01$). Thus, temperature perception was manipulated successfully while keeping possible confounders constant.

7.2.2. Perceived newness and product evaluation

Consistent with the previous experiments, participants in the cold condition perceived the product as newer than did those in the warm condition ($M_{warm} = 3.25$, $SD_{warm} = 1.04$ vs. $M_{cold} = 3.69$, $SD_{cold} = 1.16$; $F(1, 219) = 8.56$, $p < .01$, $\eta_p^2 = .04$), replicating H1a.

We also investigated the downstream effect of newness perception on product evaluation. We ran a multiple-regression analysis with perceived newness as the independent variable, product evaluation as the dependent variable, and temporal distance (0: near, 1: distant) as the moderator, using PROCESS Model 1 (Hayes, 2021). The results showed that the main effect of perceived newness significantly influenced product evaluation ($B = 0.14$, $SE = 0.07$, $t = 2.15$, $p = .03$), whereas the main effect of temporal distance was non-significant ($B = 0.11$, $SE = 0.15$, $t = 0.75$, $p = .46$). Importantly, the interaction between perceived newness and temporal distance was significant ($B = 0.74$, $SE = 0.13$, $t = 5.74$, $p < .01$). A spotlight analysis revealed that in the distant-future condition, perceived newness positively influenced product evaluation ($B = 0.53$, $SE = 0.10$, $t = 5.53$, $p < .01$). However, this effect was

reversed in the near-future condition (B = -0.22, SE = 0.09, t = -2.48, p = .01).

7.2.3. Moderated mediation

To test H3, we performed a moderated mediation analysis with background ad imagery (0: warm, 1: cold; dummy coded) as the independent variable, product evaluation as the dependent variable, temporal distance (0: near, 1: distance; dummy coded) as the moderator, and perceived newness as the mediator, using PROCESS Model 14 (Hayes, 2021) (Fig. 4). The analysis revealed significant relationships between background ad imagery and perceived newness (B = 0.44, SE = 0.15, t = 2.93, p < .01), and perceived newness and product evaluation (B = 0.15, SE = 0.07, t = 2.29, p = .02). The direct effect of background ad imagery on product evaluation was non-significant (B = -0.14, SE = 0.15, t = -0.91, p = .36). On the contrary, as reported in the preceding subsection, the interaction between perceived newness and temporal distance was significant (B = 0.76, SE = 0.13, t = 5.81, p < .01).

Moreover, the bootstrapping with 10,000 resamples revealed a significant moderated mediation effect (index = 0.33, SE = 0.14, 95% CI [0.10, 0.63]). In the distant-future condition, the indirect effect of perceived newness was positively significant (index = 0.52, SE = 0.24, 95% CI [0.07, 0.45]), whereas the indirect effect was negatively significant in the near-future condition (index = -0.09, SE = 0.05, 95% CI [-0.22, -0.01]). These results support H3 that when planning a temporally distant (vs. near) future purchase, perceived newness enhanced by background ad imagery improves (vs. impairs) consumers' evaluation of the advertised product.

7.2.4. Replication

Additional analysis using PROCESS Model 6 revealed that (1) background ad imagery associated with coldness (vs. warmth) enhanced perceived newness of the product (B = 0.44, SE = 0.14, t = 3.19, p < .01) and (2) this effect was sequentially mediated by psychological distance and ambiguity (index = 0.09, SE = 0.04, 95% CI [0.02, 0.19]), replicating H1b. Consistent with Study 3, participants' mood status did not differ between the warm and cold conditions. The full results of these additional analyses are reported in the Online Supplemental Material.

8. General discussion

Through four experiments, we examined how the perceived temperature of background ad imagery influences consumers' perception of the advertised product's newness. We demonstrated that background ad imagery associated with coldness tended to enhance the perceived newness of the advertised product (Study 1). This effect was sequentially mediated by psychological distance and ambiguity (Study 2), and manifested when the product style was modern and not antique (Study 3). Moreover, perceived newness produced by background ad imagery improved consumers' evaluation of the advertised product when they planned to purchase the product in the distant future, as opposed to impaired evaluation when the purchase was planned in the near future (Study 4). The effect of background ad imagery associated with coldness (vs. warmth) on newness perception was replicated across all studies employing different types of products and advertisements, confirming the robustness of our findings.

8.1. Theoretical contributions

The present study contributes to the literature on advertising, especially background ad imagery. Previous studies have shown that background ad imagery can convey meaningful concepts such as luxuriousness (Van Rompay et al., 2012) and greenness (Hartmann & Apaolaza-Ibáñez, 2009) of advertised products. Moreover, although extant research has also documented the effects of ad imagery on consumers' psychological distance and construal levels (e.g., Kim et al., 2019), it was unclear how ad imagery could affect these constructs. Drawing on the concept of vision–temperature correspondence, our study contributes to providing a new theoretical explanation by demonstrating the role of perceived temperature in the effect of background ad imagery on psychological distance and perceived ambiguity.

This theoretical development also has implications for the literature on product management. Although ambiguity has been considered a primary driver determining perceived newness (Blake et al., 1973; Coskuner-Balli & Sandikci, 2014; Elsbach & Stigliani, 2019), few studies have examined which factors increase/decrease the perceived ambiguity of the product. From the CLT perspective, our study contributes to filling this gap by exploring the role of psychological distance as a determinant of perceived ambiguity of a product. More broadly, this is the first study to offer a new framework explaining the psychological mechanism of newness perception by integrating different strands of research, such as vision–temperature correspondence in the advertising literature and newness perception in the product management literature.

Finally, our findings also contribute to the marketing literature focusing on temperatures. Relying on grounded cognition theory, previous studies have investigated how physical warmth affects consumer behavior. These studies have documented many positive effects on consumer judgment, such as enhancing product preference (Huang, Zhang, Hui, & Wyer, 2014) and promoting purchase behavior (Zwebner, Lee, & Goldenberg, 2014). While one study showed that coldness could in fact improve product evaluation when it comes to status signaling and luxuriousness (Park & Hadi, 2020), this was an exceptional case in a whole body of literature examining temperature's effects on consumer judgments. In the present study, we complement that exceptional finding and add nuance to the research by demonstrating that the perception of coldness (but not warmth) can improve product evaluation via perceived newness when planning a product purchase in the distant future (Study 4).

8.2. Managerial implications

Our findings also provide beneficial implications for practitioners, particularly for marketers in industries where product novelty tends to provide a competitive advantage (e.g., appliances, automobiles). As

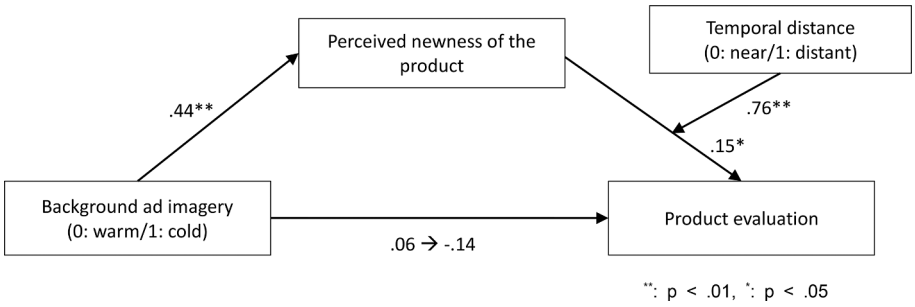


Fig. 4. Results of the moderated mediation analysis. Although the total effect (background ad imagery → product evaluation) was non-significant, this does not mean that the indirect effect cannot be examined in mediation analysis. This is because we did not predict the effect of background ad imagery on product evaluation. Additionally, a body of statistical literature argues that the significant total effect is not a prerequisite for examining indirect effects (e.g., Shrout & Bolger, 2002; Zhao, Lynch, & Chen, 2010).

newness perception is a vital selling point and a critical source of competitive advantage (Bloch, 1995; Radford & Bloch, 2011), it is imperative for marketers to consider how they communicate product newness to their target customers. Although it is possible to convey product newness by improving the product itself, this strategy requires considerable resources to redesign products, develop innovative technologies, and restructure production processes. The present study suggests an alternative approach, namely using visual advertisement elements to convey a product's newness. Specifically, marketers can take advantage of background ad imagery associated with coldness (e.g., using a photo depicting winter scenery) to communicate product newness to customers. This strategy would be more practical and effective, considering that altering advertising images has a lower cost compared to altering products (Peracchio & Meyers-Levy, 2005), and using images has become increasingly important in both traditional, offline media and new, online media (Leonhardt, Catlin, & Pirouz, 2015). Indeed, major automobile manufacturers, such as Toyota and Mercedes-Benz, depict their futuristic concept cars with bluish, wintry imagery (see Mercedes-Benz, 2020; Toyota, 2019 for images). Additionally, our findings highlight the importance of considering temporal distance to product purchases when using background ad imagery to improve product preferences. For example, when a product will be launched in the distant future, marketers should communicate the product's newness by using images related to coldness (vs. warmth) in their advertisements.

8.3. Limitations and future research

While our findings have theoretical and managerial implications, they contain several limitations. First, the empirical validity of the results can be reinforced further. Although the underlying mechanism of the effect of background ad imagery on perceived newness was established in Studies 2 and 4, Study 3 did not examine how the temporality of product style moderated the relationship between the mediators. In other words, the mediators were not measured. Thus, to ensure empirical validity, further research testing the entire model that includes all moderators and mediators should be conducted.

Second, we used only limited types of visual cues across the experiments. We employed the content of background ad imagery to manipulate the perceived temperature in all the experiments based on the examples of actual advertisements listed in the previous section. However, vision–temperature correspondence can also be observed by employing other visual manipulations (Ho, Van Doorn, Kawabe, Watanabe, & Spence, 2014). Future research may, for example, focus on color (e.g., red vs. blue), which is strongly tied to the perception of temperature (Szocs & Biswas, 2013).

Third, across the experiments, we recruited participants from a single country, Japan. Thus, future research should examine cultural differences in the results of this study. Psychological literature shows that people in Eastern (vs. Western) countries tend to have a holistic (vs. analytic) processing style and pay great attention to the background (vs. focal objects) of a picture (Masuda & Nisbett, 2001). Given this finding, the effect of background ad imagery associated with coldness on newness perception may be more prominent in Eastern (vs. Western) countries.

Finally, considering our research aim, we focused only on coldness as a predictor of newness perception. Therefore, our results prompt a new question, that is, whether warmth is associated with the perception of oldness. We did not include this question because it is beyond the scope of our research objectives. Thus, future research can examine whether background ad imagery typically associated with warmth (e.g., a photo depicting a fireplace), generates past-related perceptions (e.g., nostalgia), and conveys the product's oldness.

8.4. Conclusions

This study investigated the influence of vision–temperature correspondence on newness perception and product evaluation. Through the experiments in this study, we successfully demonstrated that background ad imagery associated with coldness enhances the perceived newness of advertised products. The effect is sequentially mediated by psychological distance and ambiguity, and it manifests when the product style is modern, not antique. Additionally, the effect has been seen to improve product evaluation when the purchase is considered for the distant future.

CRediT authorship contribution statement

Taku Togawa: Project administration, Investigation, Funding acquisition, Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Hiroaki Ishii:** Writing – review & editing, Methodology, Conceptualization. **Jaewoo Park:** Writing – review & editing, Methodology, Conceptualization. **Rajat Roy:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the associate editor and anonymous reviewers for their helpful and constructive comments provided throughout the review process.

Funding

This work was supported by the Japan Society for the Promotion of Science (Grant-in-Aid for Scientific Research, grant number: 19K01943); the opinions expressed here are our own and do not necessarily reflect the views of the funding organization.

Appendix A. Stimuli used in Study 1



Warm condition



Cold condition

Appendix B. Stimuli used in Study 2



Warm condition



Cold condition

Appendix C. Stimuli used in Study 3



Warm/modern



Cold/modern



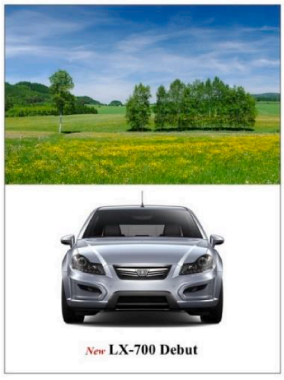
Warm/antique



Cold/antique

NOTE. The advertisement copy on the upper-right side of each stimulus means "Good Furniture, Good Life" in Japanese.

Appendix D. Stimuli used in Study 4



Warm condition



Cold condition

Appendix E. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbusres.2023.113801.

References

Aaker, J. L. (2000). Accessibility or diagnosticity? Disentangling the influence of culture on persuasion processes and attitudes. *Journal of Consumer Research, 26*(4), 340–357. https://doi.org/10.1086/209567

Barsalou, L. W. (2010). Grounded cognition: Past, present, and future. *Topics in Cognitive Science, 2*(4), 716–724. https://doi.org/10.1111/j.1756-8765.2010.01115.x

Bartels, J., & Reinders, M. J. (2011). Consumer innovativeness and its correlates: A propositional inventory for future research. *Journal of Business Research, 64*(6), 601–609. https://doi.org/10.1016/j.jbusres.2010.05.002

Biliciler, G., Raghunathan, R., & Ward, A. F. (2021). Consumers as naive physicists: How visual entropy cues shift temporal focus and influence product evaluations. *Journal of Consumer Research, 48*(6), 1010–1031. https://doi.org/10.1093/jcr/ucab042

Blake, B. F., Perloff, R., Zenhausern, R., & Heslin, R. (1973). The effect of intolerance of ambiguity upon product perceptions. *Journal of Applied Psychology, 58*(2), 239–243. https://doi.org/10.1037/h0035651

Bloch, P. H. (1995). Seeking the ideal form: Product design and consumer response. *Journal of Marketing, 59*(3), 16–29. https://doi.org/10.2307/1252116

Blythe, J. (1999). Innovativeness and newness in high-tech consumer durables. *Journal of Product & Brand Management, 8*(5), 415–429. https://doi.org/10.1108/10610429910296028

Calantone, R. J., Chan, K., & Cui, A. S. (2006). Decomposing product innovativeness and its effects on new product success. *Journal of Product Innovation Management, 23*(5), 408–421. https://doi.org/10.1111/j.1540-5885.2006.00213.x

Castagna, A. C., Pinto, D. C., Mattila, A., & de Barcellos, M. D. (2021). Beauty-is-good, ugly-is-risky: Food aesthetics bias and construal level. *Journal of Business Research, 135*, 633–643. https://doi.org/10.1016/j.jbusres.2021.06.063

Chae, B., & Hoegg, J. (2013). The future looks "right": Effects of the horizontal location of advertising images on product attitude. *Journal of Consumer Research, 40*(2), 223–238. https://doi.org/10.1086/669476

Cho, Y.-N., & Taylor, C. R. (2020). The role of ambiguity and skepticism in the effectiveness of sustainability labeling. *Journal of Business Research, 120*, 379–388. https://doi.org/10.1016/j.jbusres.2019.08.034

Choi, J. D., Rangan, P., & Singh, S. N. (2016). Do cold images cause cold-heartedness? The impact of visual stimuli on the effectiveness of negative emotional charity appeals. *Journal of Advertising, 45*(4), 417–426. https://doi.org/10.1080/00913367.2016.1185982

Clark, L. A., & Watson, D. (1989). *The Japanese Positive and Negative Affect Schedule: Factor-based scales for the assessment of mood [Unpublished manuscript]*. University of Iowa.

Coskuner-Balli, G., & Sandikci, Ö. (2014). The aura of new goods: How consumers mediate newness. *Journal of Consumer Behaviour, 13*(2), 122–130. https://doi.org/10.1002/cb.1470

Creusen, M. E. H., & Schoormans, J. P. L. (2005). The different roles of product appearance in consumer choice. *Journal of Product Innovation Management, 22*(1), 63–81. https://doi.org/10.1111/j.0737-6782.2005.00103.x

Dahl, D. W., & Moreau, P. (2002). The influence and value of analogical thinking during new product ideation. *Journal of Marketing Research, 39*(1), 47–60. https://doi.org/10.1509/jmkr.39.1.47.18930

Elsbach, K. D., & Stigliani, I. (2019). New information technology and implicit bias. *Academy of Management Perspectives, 33*(2), 185–206. https://doi.org/10.5465/amp.2017.0079

Ernst, M. O., & Banks, M. S. (2002). Humans integrate visual and haptic information in a statistically optimal fashion. *Nature, 415*(6870), 429–433. https://doi.org/10.1038/415429a

Fujita, K., Eyal, T., Chaiken, S., Trope, Y., & Liberman, N. (2008). Influencing attitudes toward near and distant objects. *Journal of Experimental Social Psychology, 44*(3), 562–572. https://doi.org/10.1016/j.jesp.2007.10.005

George, J. M., & Zhou, J. (2002). Understanding when bad moods foster creativity and good ones don't: The role of context and clarity of feelings. *Journal of Applied Psychology, 87*(4), 687–697. https://doi.org/10.1037/0021-9010.87.4.687

Goode, M. R., Dahl, D. W., & Moreau, C. P. (2013). Innovation aesthetics: The relationship between category cues, categorization certainty, and newness perceptions. *Journal of Product Innovation Management, 30*(2), 192–208. https://doi.org/10.1111/j.1540-5885.2012.00995.x

Gvili, Y., Tal, A., Amar, M., & Wansink, B. (2017). Moving up in taste: Enhanced projected taste and freshness of moving food products. *Psychology & Marketing, 34*(7), 671–683. https://doi.org/10.1002/mar.21014

Hagtvedt, H., & Patrick, V. M. (2008). Art infusion: The influence of visual art on the perception and evaluation of consumer products. *Journal of Marketing Research, 45*(3), 379–389. https://doi.org/10.1509/jmkr.45.3.379

Hartmann, P., & Apaolaza-Ibáñez, V. (2009). Green advertising revisited. *International Journal of Advertising, 28*(4), 715–739. https://doi.org/10.2501/S0265048709200837

Hayes, A. F. (2021). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach* (3rd ed.). Guilford Press.

Hetet, B., Ackermann, C. L., & Mathieu, J. P. (2020). The role of brand innovativeness on attitudes towards new products marketed by the brand. *Journal of Product and Brand Management, 29*(5), 569–581. https://doi.org/10.1108/JPBM-02-2019-2243

Hirschman, E. C. (1980). Innovativeness, novelty seeking, and consumer creativity. *Journal of Consumer Research, 7*(3), 283–295. https://doi.org/10.1086/208816

Ho, H. N., Van Doorn, G. H., Kawabe, T., Watanabe, J., & Spence, C. (2014). Colour-temperature correspondences: When reactions to thermal stimuli are influenced by colour. *PLoS ONE, 9*(3), 1–7. https://doi.org/10.1371/journal.pone.0091854

Hoch, S. J., & Ha, Y.-W. (1986). Consumer learning: Advertising and the ambiguity of product experience. *Journal of Consumer Research, 13*(2), 221–233. https://doi.org/10.1086/209062

Holbrook, M. B. (1981). Integrating compositional and decompositional analyses to represent the intervening role of perceptions in evaluative judgments. *Journal of Marketing Research, 18*(1), 13–28. https://doi.org/10.2307/3151310

Howarth, E., & Hoffman, M. S. (1984). A multidimensional approach to the relationship between mood and weather. *British Journal of Psychology, 75*(1), 15–23. https://doi.org/10.1111/j.2044-8295.1984.tb02785.x

Huang, X., Zhang, M., Hui, M. K., & Wyer, R. S. (2014). Warmth and conformity: The effects of ambient temperature on product preferences and financial decisions. *Journal of Consumer Psychology, 24*(2), 241–250. https://doi.org/10.1016/j.jcps.2013.09.009

Ijzerman, H., & Semin, G. R. (2009). The thermometer of social relations: Mapping social proximity on temperature. *Psychological Science, 20*(10), 1214–1220. https://doi.org/10.1111/j.1467-9280.2009.02434.x

Johnson, M. K., Foley, M. A., Suengas, A. G., & Raye, C. L. (1988). Phenomenal characteristics of memories for perceived and imagined autobiographical events. *Journal of Experimental Psychology: General, 117*(4), 371–376. https://doi.org/10.1037/0096-3445.117.4.371

Kanaya, S., Matsushima, Y., & Yokosawa, K. (2012). Does seeing ice really feel cold? Visual-thermal interaction under an illusory body-ownership. *PLoS ONE, 7*(11), 1–7. https://doi.org/10.1371/journal.pone.0047293

Kim, D., Hyun, H., & Park, J. (2020). The effect of interior color on customers' aesthetic perception, emotion, and behavior in the luxury service. *Journal of Retailing and Consumer Services, 57*(July), Article 102252. https://doi.org/10.1016/j.jretconser.2020.102252

Kim, K., Zhang, M., & Li, X. (2008). Effects of temporal and social distance on consumer evaluations. *Journal of Consumer Research, 35*(4), 706–713. https://doi.org/10.1086/592131

Kim, K., Lee, S., & Choi, Y. K. (2019). Image proximity in advertising appeals: Spatial distance and product types. *Journal of Business Research, 99*, 490–497. https://doi.org/10.1016/j.jbusres.2017.08.031

Landau, M. J., Meier, B. P., & Keefer, L. A. (2010). A metaphor-enriched social cognition. *Psychological Bulletin, 136*(6), 1045–1067. https://doi.org/10.1037/a0020970

Lee, H., Fujita, K., Deng, X., & Unnava, H. R. (2017). The role of temporal distance on the color of future-directed imagery: A construal-level perspective. *Journal of Consumer Research, 43*(5), 707–725. https://doi.org/10.1093/jcr/ucw051

Leonhardt, J. M., Catlin, J. R., & Pirouz, D. M. (2015). Is your product facing the ad's center? Facing direction affects processing fluency and ad evaluation. *Journal of Advertising, 44*(4), 315–325. https://doi.org/10.1080/00913367.2015.1048911

Liberman, N., Sagristano, M. D., & Trope, Y. (2002). The effect of temporal distance on level of mental construal. *Journal of Experimental Social Psychology, 38*(6), 523–534. https://doi.org/10.1016/S0022-1031(02)00535-8

Liberman, N., & Trope, Y. (1998). The role of feasibility and desirability considerations in near and distant future decisions: A test of temporal construal theory. *Journal of Personality and Social Psychology, 75*(1), 5–18. https://doi.org/10.1037/0022-3514.75.1.5

Liberman, N., & Trope, Y. (2008). The psychology of transcending the here and now. *Science, 322*(5905), 1201–1205. https://doi.org/10.1126/science.1161958

Liviatan, I., Trope, Y., & Liberman, N. (2008). Interpersonal similarity as a social distance dimension: Implications for perception of others' actions. *Journal of Experimental Social Psychology, 44*(5), 1256–1269. https://doi.org/10.1016/j.jesp.2008.04.007

Loken, B., & Ward, J. (1990). Alternative approaches to understanding the determinants of typicality. *Journal of Consumer Research, 17*(2), 111–126. https://doi.org/10.1086/208542

Ma, Z., Yang, Z., & Mourali, M. (2014). Consumer adoption of new products: Independent versus interdependent self-perspectives. *Journal of Marketing, 78*(2), 101–117. https://doi.org/10.1509/jm.12.0051

Masuda, T., & Nisbett, R. E. (2001). Attending holistically versus analytically: Comparing the context sensitivity of Japanese and Americans. *Journal of Personality and Social Psychology, 81*(5), 922–934. https://doi.org/10.1037/0022-3514.81.5.922

Mercedes-Benz. (2020). *Inspired by the future: The Mercedes-Benz VISION AVTR.* <https://www.mercedes-benz.com/en/vehicles/passenger-cars/mercedes-benz-concept-cars/vision-avtr/>.

Noble, C. H., & Kumar, M. (2010). Exploring the appeal of product design: A grounded, value-based model of key design elements and relationships. *Journal of Product Innovation Management, 27*(5), 640–657. https://doi.org/10.1111/j.1540-5885.2010.00742.x

Park, J., & Hadi, R. (2020). Shivering for status: When cold temperatures increase product evaluation. *Journal of Consumer Psychology, 30*(2), 314–328. https://doi.org/10.1002/jcpy.1133

Peracchio, L. A., & Meyers-Levy, J. (2005). Using stylistic properties of ad pictures to communicate with consumers. *Journal of Consumer Research, 32*(1), 29–40. https://doi.org/10.1086/429599

Radford, S. K., & Bloch, P. H. (2011). Linking innovation to design: Consumer responses to visual product newness. *Journal of Product Innovation Management, 28*(SUPPL. 1), 208–220. https://doi.org/10.1111/j.1540-5885.2011.00871.x

Roehm, M. L., & Roehm, H. A. (2010). The influence of redemption time frame on responses to incentives. *Journal of the Academy of Marketing Science, 38*(3), 363–375. https://doi.org/10.1007/s11747-009-0166-9

Roque, J., Auvray, M., & Lafraire, J. (2018). Understanding freshness perception from the cognitive mechanisms of flavor: The case of beverages. *Frontiers in Psychology, 8*, 2360. https://doi.org/10.3389/fpsyg.2017.02360

Sánchez, A. M., Coleman, C. W., & Ledgerwood, A. (2021). Does temporal distance influence abstraction? A large pre-registered experiment. *Social Cognition, 39*(3), 352–365. https://doi.org/10.1521/soco.2021.39.3.352

Schnurr, B. (2017). The impact of atypical product design on consumer product and brand perception. *Journal of Brand Management, 24*(6), 609–621. https://doi.org/10.1057/s41262-017-0059-z

Shrout, P. E., & Bolger, N. (2002). Mediation in experimental and nonexperimental studies: New procedures and recommendations. *Psychological Methods, 7*(4), 422–445. https://doi.org/10.1037/1082-989X.7.4.422

Snefjella, B., & Kuperman, V. (2015). Concreteness and psychological distance in natural language use. *Psychological Science, 26*(9), 1449–1460. https://doi.org/10.1177/0956797615591771

Szocs, C., & Biswas, D. (2013). The effects of color on food temperature perceptions. In S. Botti, & A. Labroo (Eds.), *NA - Advances in Consumer Research* (Vol. 41, pp. 721–722). Association for Consumer Research.

Toyota. (2019). *Our vision for the car of the future starts with "Yui." The future has arrived.* <https://www.toyota.com/concept-i/>.

Trope, Y., & Liberman, N. (2000). Temporal construal and time-dependent changes in preference. *Journal of Personality and Social Psychology, 79*(6), 876–889. https://doi.org/10.1037//0022-3514.79.6.876

Trope, Y., & Liberman, N. (2003). Temporal construal. *Psychological Review, 110*(3), 403–421. https://doi.org/10.1037/0033-295X.110.3.403

Trope, Y., & Liberman, N. (2010). Construal-level theory of psychological distance. *Psychological Review, 117*(2), 440–463. https://doi.org/10.1037/a0018963

Trope, Y., Liberman, N., & Wakslak, C. (2007). Construal levels and psychological distance: Effects on representation, prediction, evaluation, and behavior. *Journal of Consumer Psychology, 17*(2), 83–95. https://doi.org/10.1016/S1057-7408(07)70013-X

Van Boven, L., Kane, J., McGraw, A. P., & Dale, J. (2010). Feeling close: Emotional intensity reduces perceived psychological distance. *Journal of Personality and Social Psychology, 98*(6), 872–885. https://doi.org/10.1037/a0019262

Van Rompay, T. J. L., De Vries, P. W., Bontekoe, F., & Tanja-Dijkstra, K. (2012). Embodied product perception: Effects of verticality cues in advertising and packaging design on consumer impressions and price expectations. *Psychology & Marketing, 29*(12), 919–928. https://doi.org/10.1002/mar.20574

Williams, L. E., & Bargh, J. A. (2008). Experiencing physical warmth promotes interpersonal warmth. *Science, 322*(5901), 606–607. https://doi.org/10.1126/science.1162548

Wohl, M. J. A., & McGrath, A. L. (2007). The perception of time heals all wounds: Temporal distance affects willingness to forgive following an interpersonal transgression. *Personality and Social Psychology Bulletin, 33*(7), 1023–1035. https://doi.org/10.1177/0146167207301021

Yan, D., & Sengupta, J. (2011). Effects of construal level on the price-quality relationship. *Journal of Consumer Research, 38*(2), 376–389. https://doi.org/10.1086/659755

Zhang, M., & Wang, J. (2009). Psychological distance asymmetry: The spatial dimension vs. other dimensions. *Journal of Consumer Psychology, 19*(3), 497–507. https://doi.org/10.1016/j.jcps.2009.05.001

Zhao, M., & Xie, J. (2011). Effects of social and temporal distance on consumers' responses to peer recommendations. *Journal of Marketing Research, 48*(3), 486–496. https://doi.org/10.1509/jmkr.48.3.486

Zhao, X., Lynch, J. G., & Chen, Q. (2010). Reconsidering Baron and Kenny: Myths and truths about mediation analysis. *Journal of Consumer Research, 37*(2), 197–206. https://doi.org/10.1086/651257

Zwebner, Y., Lee, L., & Goldenberg, J. (2014). The temperature premium: Warm temperatures increase product valuation. *Journal of Consumer Psychology, 24*(2), 251–259. https://doi.org/10.1016/j.jcps.2013.11.003

Dr. Taku Togawa is an Associate Professor of Marketing at Sophia University, Japan. He mainly conducts experimental studies in the area of consumer behavior. Specifically, his research focuses on how and when sensory inputs from marketing tools, including advertising, packaging, and in-store environments, influence consumers' perceptions, evaluations, and consumption behavior. He has published in academic journals such as the *Journal of Consumer Psychology*, *Journal of Retailing*, *Marketing Intelligence and Planning*, *Food Quality and Preference*, *Personality and Social Psychology Bulletin*, and *Motivation Science*.

Dr. Hiroaki Ishii is an Associate Professor of Marketing at Aoyama Gakuin University, Japan. His research focuses on consumer psychology, brand management, marketing communications, and sensory marketing. He is particularly interested in the interaction effects of sensory and other factors generated by advertising, packaging, and in-store promotion. His studies have been published in the *Journal of Retailing*, *Psychology & Marketing*, *Marketing Intelligence and Planning*, and others.

Dr. Jaewoo Park is a Professor of Marketing at Chuo University, Japan. His research interests include sensory marketing, design aesthetics, luxury perception, and branding. His scholarly work has been published in academic journals such as the *Journal of Consumer Psychology*, *Journal of Retailing*, *Journal of Advertising*, *Journal of Business Research*, *Psychology & Marketing*, *Journal of Product and Brand Management*, and *Food Quality and Preference*.

Dr. Rajat Roy is an Associate Professor of Marketing at Bond University, Australia. His predominant research interest lies in consumer behavior. He has published in the *Journal of Consumer Psychology*, *Journal of Advertising*, *Journal of Business Research*, *Psychology & Marketing*, and *European Journal of Marketing*, among others. He has undertaken cross-disciplinary research with health sciences and published in *PLOS One*. He sits on the editorial boards of the *Journal of Business Research*, *International Journal of Advertising*, *European Journal of Marketing*, and *Journal of Advertising Research*. He serves as an Associate Editor for the *Journal of Creative Communications*. He has consulted for industry clients such as Mercedes Benz and the United States Studies Centre, obtained Australian and international grants, and contributed to the Australian Federal Government's policy paper on research commercialization in Australian universities. He was acknowledged as one of the top 60 global scholars for his contribution to advertising research.